Do Now · retrieval

Lesson 107 · Paging, Segmentation and Virtual Memory

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Identify two services provided by an OS.

Q2

What is the time complexity of sequential search in the worst case?

Q3

After pushing A then B onto a stack, which is popped first?

Q4

Why must a binary search operate on ordered data?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Paging — Virtual memory is split into fixed-size pages; RAM into frames.

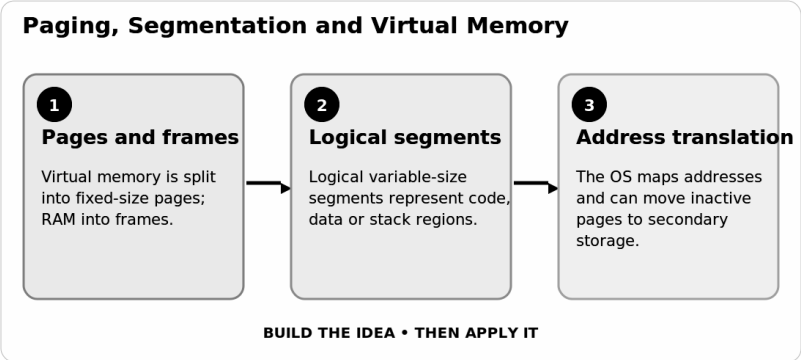
Core idea 2

Segmentation — Logical variable-size segments represent code, data or stack regions.

Core idea 3

Virtual memory — The OS maps addresses and can move inactive pages to secondary storage.

Key visual · complete concept



New Learning · applying the ideas

Vocabulary · use the correct technical terms

page — Fixed-size virtual-memory block.

frame — Fixed-size physical-RAM allocation unit.

segmentation — Memory organised into variable-size logical segments.

page fault — Access to a page not currently resident in RAM.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

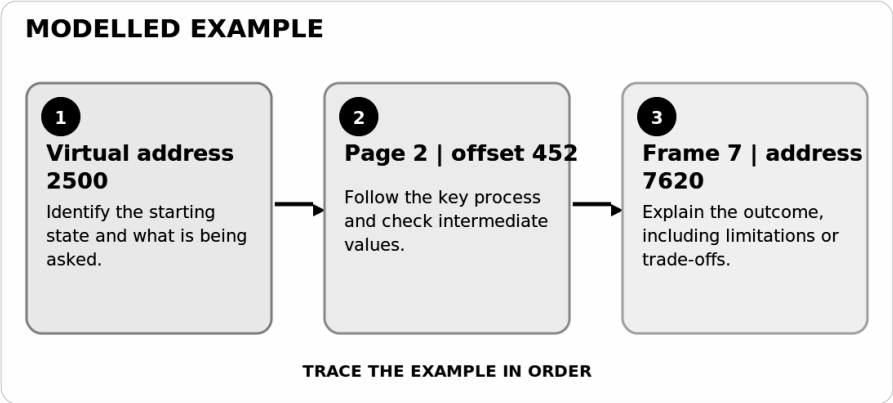
Problem

A 2,500-byte virtual address uses 1,024-byte pages. Find the page and offset, then place it in frame 7.

Follow the worked method

1. Model answer — Page number = 2 and offset = 452. Frame 7 base is 7 × 1,024 = 7,168; physical address = 7,620, assuming that page is mapped to frame 7. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Differentiate paging and segmentation, with one reason for using each. [6]

Question 2

With 4,096-byte pages, calculate page number and offset of virtual address 8,193. [4]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why heavy paging can cause thrashing and slow down a computer. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Virtual memory increases the speed of every memory access.

Use key vocabulary from memory

1. Explain page in your own words.

2. Explain frame in your own words.

3. Explain segmentation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.